Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
9.	(a)			mass of NO_2 = 25.0 × 0.714 ÷ 100 = 0.1785 (1) moles NO_2 = 0.1785 ÷ 46 = 3.88 × 10^{-3} (1) answer must be to 3 sig figs to gain full marks		2		2	2	
	(b)	(i)		$K_c = \frac{[NO_2]^2}{[N_2O_4]}$ (1) unit mol dm^{-3} (1) allow ecf for unit		2		2	1	
		(ii)	I	concentration of NO_2 = 2 × (0.400 – 0.0581) = 0.6838 (1) value of K_c = 8.05 (1) solvent is CCl_4 (1) (allow ecf)		2	1	3	1	
			II	CS_2 given with any valid reason (1) explanation in terms of position of equilibrium or amount of product formed (1)		1	1	2		